

Tharun Prasath J

Ph.D. Scholar, Department of Electrical Engineering, IIT Madras, Chennai, India
tharunprasath3@gmail.com | Website | LinkedIn

Research Interests

Vision-Language Models, Gaze Tracking, Egocentric Video Analysis, Temporal Sequence Modeling

Education

Indian Institute of Technology Madras (IITM)

Ph.D. in Electrical Engineering

Thesis: Multimodal AI for Egocentric Video Analysis (Ongoing)

Chennai, India

July 2026 – Present

Advisor: Dr. Kaushik Mitra

Indian Institute of Space Science and Technology (IIST)

M.Tech in Digital Signal Processing (CGPA: 9.51 / 10.0)

Thesis: From Descriptors to Manifolds: A Progressive Framework for Ocular Emotion Recognition Using Gaze and Pupil Dynamics

Thiruvananthapuram, India

2024 – 2026

Advisor: Dr. Deepak Mishra

St. Joseph's College of Engineering

B.E. in Electronics and Communication Engineering (CGPA: 9.48 / 10.0)

Chennai, India

2019 – 2023

Publications & Preprints

- **EmotionMamba: Event-Conditioned State-Space Modeling of Gaze and Pupil Dynamics for Emotion Recognition**
Tharun Prasath J, Deepak Mishra. *Under 1st Revision*, IEEE Transactions on Affective Computing (TAC).
- **Emotion Recognition from Eye Dynamics: Short-Window Descriptors with Gradient-Boosted Trees**
Tharun Prasath J, Deepak Mishra. *Under 2nd Revision*, IEEE Transactions on Cognitive and Developmental Systems (TCDS).
- **Spiking-Mamba: Neuromorphic Emotion Recognition from Eye-Tracking via Selective State-Space Models**
Tharun Prasath J, Deepak Mishra. *Under Review*, IEEE Transactions on Cognitive and Developmental Systems (TCDS).
- **Investigating the Current Trends of IoT on Indian Agriculture**
Tharun Prasath J, M. P. Rajakumar. *Accepted*, India Internet Governance Forum (IIGF), United Nations IGF.

Selected Projects

- **Ocular Emotion Recognition:** Developed state-space (Mamba), spiking neural, and tree-based models on gaze and pupil dynamics for real-time affective computing.
- **Virtual Fence:** Built a real-time geofencing maritime safety system for fishermen at the India–Sri Lanka maritime border.
- **Smart Elector Validation System:** Designed an offline-capable biometric authentication pipeline for polling booths.
- **Uzhavu:** Developed an optimized, direct-to-retail agricultural supply-chain pipeline for local farming communities.

Teaching Experience

Graduate Teaching Assistant – Indian Institute of Space Science and Technology (IIST)

- **Computer Vision (PG Course):** Assisted with coursework, evaluations, and student mentoring.
- **Machine Learning for Signal Processing (MLSP) Lab:** Conducted hands-on coding and lab practicals.

Honors & Achievements

- **First Prize, PG Day Best Project (IIST):** Awarded best M.Tech thesis project across the institute.
- **IndiaAI Fellowship Scheme:** Awarded INR 2 Lakhs fellowship by MeitY, Govt. of India, for AI project development.
- **FAER Scholar Award (2022–23):** Received final-year project development grant from FAER, Bengaluru.
- **Optica Student Leadership Conference (USA):** Selected among the Top 100 students globally (New York, USA).
- **National Hackathons:** 3rd Place in Sony–IITM Samvedan (2021); 3rd Place in SERB–INAE Conclave (IIT Kanpur, 2022); Finalist in Bharatiya Antariksh Hackathon (NRSC/ISRO), AWS Innovation, and HackofPi.

Technical Skills

- **Programming & Tools:** Python, Linux/Bash, Git, L^AT_EX
- **ML/DL Frameworks:** PyTorch, Hugging Face, Scikit-learn, OpenCV
- **Hardware:** Tobii Pro Glasses 3 Eye Tracker